



Article

IoT-Enabled Soil Moisture and Conductivity Monitoring Under Controlled and Field Fertigation Systems

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Abstract

Precision agriculture increasingly relies on real-time data from soil sensors to optimize irrigation and nutrient application. Soil moisture and electrical conductivity (EC) are key indicators in irrigation and fertigation systems, directly affecting water-use efficiency and nutrient delivery to crops. This study evaluates the performance of an IoT-based soil-monitoring system for real-time tracking of EC and soil moisture under varied fertigation conditions in both laboratory and field scenarios. The EC sensor showed strong agreement with laboratory YSI measurements ($R^2 = 0.999$), confirming its accuracy. Column experiments were conducted in three soil types (sand, sandy loam, and loamy sand) to assess the EC and soil moisture response to fertigation. Sand showed rapid infiltration and low retention, with EC peaking at $420 \mu\text{S}/\text{cm}$ and moisture $0.33 \text{ cm}^3/\text{cm}^3$, indicating high leaching risk. Sandy loam retained the most moisture ($0.35 \text{ cm}^3/\text{cm}^3$) and showed the highest EC ($550 \mu\text{S}/\text{cm}$), while loamy sand exhibited intermediate behavior. Fertilizer-specific responses showed higher EC in Calcium Ammonium Nitrate (CAN)-treated soils, while Monoammonium Phosphate (MAP) showed lower, more stable EC due to limited phosphorus mobility. Field validation confirmed that the IoT system effectively captured irrigation and fertigation events through synchronized EC and moisture peaks. These findings highlight the efficacy of IoT-based sensor networks for continuous, high-resolution soil monitoring and their potential to support precision fertigation strategies, enhancing nutrient-use efficiency while minimizing environmental losses.

Keywords: IoT-based sensors; electrical conductivity; soil moisture; fertigation; precision agriculture



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1. Introduction

Agricultural productivity and sustainability have become critical concerns in modern farming practices, especially in the context of precision agriculture [1]. In modern agriculture, fertigation refers to the combined application of fertilizers and irrigation. It has become a cornerstone of precision farming, offering targeted nutrient delivery that enhances crop yield and resource efficiency [2]. However, the efficacy of fertigation systems is often compromised by mechanical failures, leading to improper nutrient dosing that can adversely affect plant health and contribute to environmental degradation through nutrient leaching and runoff [3]. Therefore, ensuring the efficiency and reliability of fertigation systems is paramount to maintaining sustainable agricultural practices.

Soil electrical conductivity (EC) is a crucial parameter in agricultural soil management, providing insights into soil salinity, nutrient availability, and water retention character-

istics [4]. EC can serve as an indirect indicator of dissolved ions such as cations and anions [4]. Monitoring EC can facilitate real-time assessments of soil fertility and help identify potential risks of nutrient leaching [5]. Among soil types, sandy soils specifically pose unique challenges in terms of nutrient retention and water management due to their high permeability and low cation exchange capacity (CEC), making it particularly difficult to manage effectively [6]. Excessive nutrient leaching in such soil types is a significant environmental concern as it can lead to groundwater contamination, eutrophication, and public health risks such as methemoglobinemia [7]. For instance, nitrate is an essential nutrient for plant growth but is highly susceptible to leaching due to its solubility and weak adsorption onto soil particles [8]. Unlike other macronutrients such as phosphorus, which tends to bind to soil particles, nitrate remains predominantly in the soil solution, making its movement largely dependent on water flow [9]. Because EC is a measure of the total dissolved ion concentration in the soil solution, it has been used as a surrogate indicator for nitrate monitoring [10]. Several studies have demonstrated that soil EC correlates well with nitrate concentration in both laboratory and field conditions [4,7,11,12]. However, this relationship is not always linear due to the presence of other dissolved ions such as sulfate, chloride, and bicarbonates, which also contribute to EC variations [10].

In parallel to soil EC, soil moisture monitoring plays a pivotal role in understanding plant-available water metrics and soil nutrient mobility, especially under these fertigation systems [13]. Soil moisture content influences solute transport, root zone dynamics, and crop response to irrigation scheduling [14]. Accurate and continuous monitoring of soil moisture allows for better irrigation management and improved water-use efficiency [15]. Various sensing technologies including capacitance, time-domain reflectometry (TDR), and resistance-based sensors have been employed to monitor moisture at different soil depths [16]. In sandy soil, rapid drainage following irrigation events lead to short retention times for both moisture and nutrients, making moisture monitoring essential for preventing nutrient loss through deep percolation [17]. Additionally, the coupling of EC and soil moisture data enables a more accurate interpretation of ion fluxes and fertilizer behavior in the soil matrix [4,18].

The emergence of Internet of Things (IoT) technologies in agriculture has significantly revolutionized soil-monitoring practices. IoT-based systems enable real-time data acquisition and remote management of soil conditions, facilitating timely interventions to maintain optimal fertigation practices [19]. These systems allow for real-time monitoring of both EC and soil moisture, significantly enhancing data-driven decision-making in agriculture. A recent study developed an IoT-based fertigation system capable of automatically applying fertilizer mixtures with consistent EC values to crops, reducing human error and increasing system efficiency [20]. By integrating soil EC and moisture sensing, IoT-enabled systems offer a comprehensive view of soil dynamics, supporting more accurate nutrient transport modeling and adaptive management strategies that enhance fertilizer efficiency and environmental sustainability. Despite growing interest in precision fertigation, the practical deployment of soil EC and moisture-monitoring systems faces several challenges, including the high cost of sensors, limited temporal resolution, lack of real-time data access, and difficulties integrating multi-parameter sensing into a unified, scalable platform [16,21]. Conventional systems often lack the responsiveness required to detect short-duration fertigation events or sudden changes in nutrient concentrations, leading to suboptimal nutrient application and potential environmental losses [19]. These limitations highlight the necessity for an integrated, cost-effective, IoT-based solution that can provide high-frequency, real-time measurements of key soil parameters to inform fertigation decisions.

The present study aims to (i) evaluate the performance and calibration accuracy of an IoT-based sensor system for real-time monitoring of soil EC and moisture content;

(ii) investigate the temporal variations in EC and soil moisture in different soil types (sand, sandy loam, and loamy sand) under fertigation conditions; (iii) compare EC and moisture dynamics in laboratory column experiments and open-field conditions for validating the sensor system's applicability. The study aims to enhance decision-making for nutrient management by providing real-time continuous data. It also contributes to precision agriculture by demonstrating how an integrated EC–moisture-monitoring system can guide fertigation scheduling to minimize nutrient losses.

2. Materials and Methods

Fertigation systems are widely used in agriculture for precise delivery of nutrients to crops but mechanical failure results in excessive nutrients supply, which negatively affects the crop yield. To prevent over-fertilization and detect system malfunction on the farm, this research implemented soil EC monitoring as a diagnostic tool. Experiments were conducted under both laboratory and field conditions, using farmers' standard fertigation practices to evaluate the effectiveness of detecting fertilizer from this EC monitoring.

2.1. IoT-Based Soil Moisture and EC-Monitoring System

2.1.1. Hardware and Software Design

For soil EC monitoring effectiveness, Particle Argon microcontrollers (San Jose, CA, USA) integrated with a custom Printed Circuit Board (PCB) were utilized to collect data from the TEROS-12 sensors. The TEROS 12 is a sensor that is designed to measure soil moisture, temperature, and EC. It features a soil moisture range of 0.00–0.70 m³/m³ for mineral soils and 0.0–1.0 m³/m³ for soilless media, with resolution as fine as 0.0010 m³/m³. Its 'generic calibration' offers an accuracy of ± 0.03 m³/m³ in mineral soils with solution EC < 8000 μ S/cm. The sensor also measures bulk EC with a range of 0–20,000 μ S/cm, a resolution of 1 μ S/cm, and an accuracy of $\pm 5\% + 10$ μ S/cm from 0–10,000 μ S/cm and $\pm 8\%$ from 10,000–20,000 μ S/cm. Additionally, the temperature measurement range extends from –40 to +60 °C with an accuracy of ± 0.3 °C when readings are within the 0 to 60 °C range.

The selection of hardware and software components for the IoT-based monitoring system was guided by their proven applicability in environmental sensing and prior use in agricultural research [4,19]. The PCB board is designed to accommodate the Particle Argon and includes a SJ-435108 TRRS plug for interfacing with the TEROS-12 sensors. Sensor communication was established via the SDI-12 standard. Soil EC data were collected periodically at 15 min intervals and sent to Ubidots, an IoT cloud platform, while also being stored locally on a microSD card for backup. The microSD card ensures reliable storage in the event of internet disruptions, while the Wi-Fi-enabled Particle Argon microcontroller supports real-time, cloud-based data access via the Ubidots cloud platform. Soil moisture was calculated from the raw sensor data using Equation (1) (MeterGroup, TEROS 11/12 manual). The physical arrangement of the IoT-based soil-monitoring system is shown in Figure 1, which displays the integration of TEROS-12 sensors with the Particle Argon microcontroller, data-logging module, and power supply components under both laboratory and field conditions. This setup enabled real-time data collection of soil EC and moisture across different soil depths and experimental treatments. To better visualize the data flow, Figure 2 presents the schematic workflow of the system. The TEROS-12 sensors communicate with the Particle Argon microcontroller via the SDI-12 protocol. The microcontroller processes the incoming sensor signals and transmits the data to the Ubidots cloud platform over Wi-Fi. A microSD card module was incorporated to store all readings locally, ensuring data redundancy in the event of network outages. Its performance has been validated in prior studies for soil moisture and EC monitoring across various soil types [4,22]. Detailed

information of each component of the IoT-based soil moisture and EC-monitoring system is shown in Table 1. This includes sensor models, quantities, unit costs, and manufacturers, providing transparency for system replication and potential scalability for field applications. The combination of reliable hardware, efficient data processing, and cloud-based visualization forms the backbone of this IoT-based fertigation-monitoring system.

$$\theta = 3.879 \times 10^{-4} \times RAW - 0.6956 \quad (1)$$

where θ is the volumetric water content/soil moisture (m^3/m^3), and RAW is the raw sensor output.

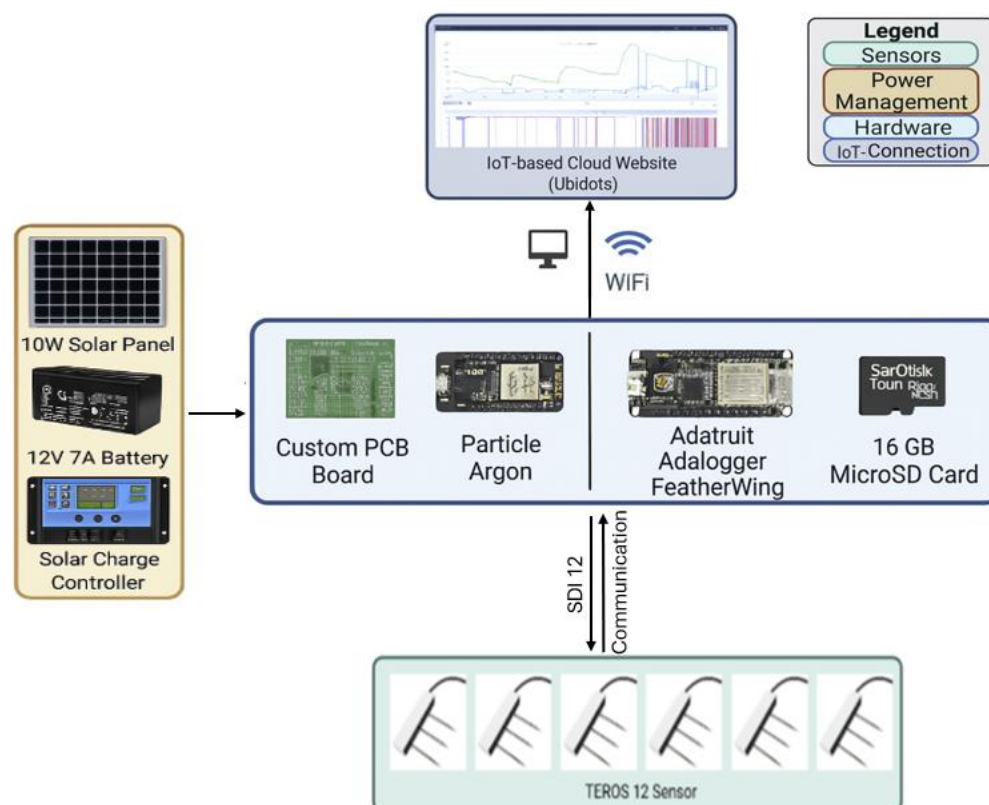


Figure 1. Experimental setup for IoT-based soil moisture and EC-monitoring system.

Table 1. List of components used in the IoT-based soil moisture and EC-monitoring system.

Components	Quantity	Cost per Unit (USD)	Manufacturer
Argon	1	27.50	Particle (San Francisco, CA, USA)
Teros 12	6	245.10	MeterGroup (Pullman, WA, USA)
Adalogger FeatherWing	1	8.95	Adafruit Industries (New York, NY, USA)
Class 10 MicroSD card	1	9.99	Sandisk (Milpitas, CA, USA)
Solar charge controller	1	29.99	EcoWorthy (NorthPoint, Hong Kong)
12V 7 Ah battery	1	20.95	Power Sonic (Reno, NV, USA)
20W 12V solar panel	1	25.00	Newpowa (Ontario, CA, USA)
ML-57F Weatherproof	1	31.59	Polycase (Avon, OH, USA)

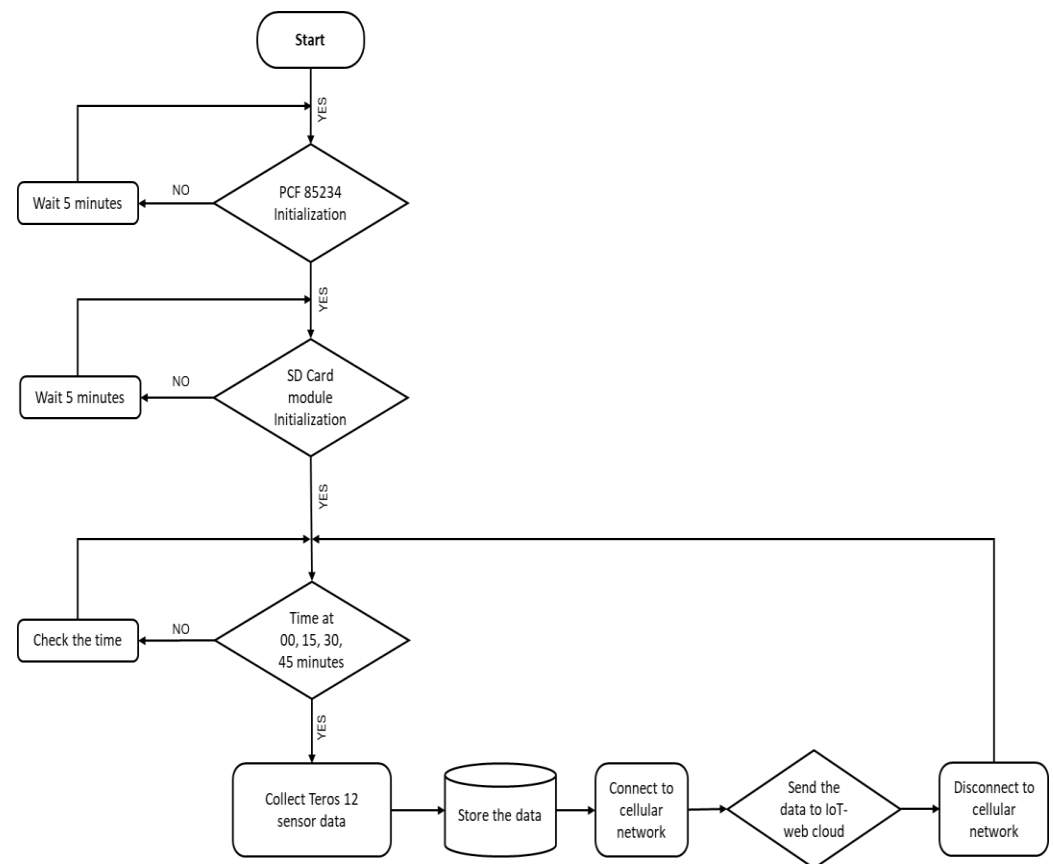


Figure 2. Schematic workflow for Particle Argon microcontroller integrated with custom PCB board (Ubidots platform) for EC monitoring.

2.1.2. Data Processing

Data from the TEROS 12 were sent to the Ubidots IoT platform. The data were transmitted from an IoT device through an API to the platform. When the data reach Ubidots, they are stored in the cloud and allow for complex processing like data aggregation, visualization, manipulation, and analysis. Two categories of data were collected from the TEROS 12 sensors: moisture content and EC.

Data cleaning is a critical step in ensuring accurate analysis and reliable results, especially when dealing with environmental data subject to noise and outliers. Soil sensor data often contains irregularities due to improper delay time between the sensor and the microcontroller resulting in improper reading. Cleaning and screening of these data was approached with caution to avoid biases to ensure the reliability of future analyses.

This approach balanced the need for outlier cleaning, which allows retaining all data points within acceptable thresholds while ensuring that meaningful fluctuations and trends are not removed or altered [23]. Outliers are replaced with rolling averages based on the 5 most recent data trends, providing a reliable and justifiable correction method. Using a rolling average, the cleaning process dynamically adapts to actual data patterns, ensuring alignment with true readings. By replacing outliers instead of removing them, it preserves dataset integrity, enabling reputable time-series analyses. This approach is adaptable and scalable, making it ideal for long-term or high-frequency datasets. The outlier removal equation is described in Equations (2) and (3).

$$\text{Cleaned value } (x_t) = \begin{cases} x_t, & \text{if } |x_t - \bar{x}_t| \leq a \\ \bar{x}_t, & \text{if } |x_t - \bar{x}_t| > a \end{cases} \quad (2)$$

where x_t refers to observed value at time t , and $\bar{x}$ is rolling average at time t (using 3 previous values) and denotes threshold value. The three-point average was calculated through Equation (3).

$$\bar{x}_t = \frac{(x_{t-3} + x_{t-2} + x_{t-1})}{3} \quad (3)$$

2.2. Sensor Calibration

To evaluate the accuracy of the IoT-based EC and moisture-monitoring system, calibration experiments were conducted prior to laboratory column study and field deployment. For EC, the sensors were calibrated against standard solutions and validated using a YSI Professional Plus multiparameter instrument (Xylem Inc., Washington, DC, USA). Calibration was performed using a series of NaCl solutions (100 to 2000 $\mu\text{S}/\text{cm}$) at 25 °C. For soil moisture, the TEROS-12 sensors were used with their factory calibration for mineral soils, which offers an accuracy of $\pm 0.03 \text{ m}^3/\text{m}^3$ for solution EC < 8000 $\mu\text{S}/\text{cm}$. Additionally, soil-specific verification was conducted in the laboratory by comparing sensor readings with gravimetric moisture content derived from oven-dried soil samples at multiple time points, validating sensor performance in sand, loamy sand, and sandy loam soils.

2.3. Experimental Design

The experimental design consisted of two phases: (i) controlled column studies and (ii) a field trial in a commercial orchard.

2.3.1. Column Studies

Soil columns were filled with three different types: sand, sandy loam, and loamy sand. TEROS-12 sensors were installed at depths of 10 and 20 cm depth in each column (Figure 3). Urea (26%) fertilizers were applied to track the changes of EC in the soil. This was replicated 4 times. The experiment comprised of plastic columns (19 Liter) filled with soil replicated three times. Each plastic column was modified by installing drainage holes with filters to ensure free percolation and simulating the water movement through the soil profile. TEROS-12 were again installed at 10 cm and 20 cm depths in the column (Figure 3). The soil was taken from the field in three layers and was kept in the column accordingly to ensure minimal disturbance and/or soil compaction. The fertigation treatment used was the common practice adopted by farmers to apple crops as 6% UAN (Urea-Ammonium Nitrate), 27.8% CaTs (Calcium Thiosulfate), and 66.2% water. The treatments comprised of common fertigation practice followed by 15 min drip irrigation (T1), fertigation and 60 min drip irrigation (T2). The soil moisture and EC data were recorded at each 15 min interval through the TEROS-12 sensor connected to the data logger. The change in EC values on fertilizer application assessed to establish the correlation between the two. The characteristics of different soils are given in Table 2.

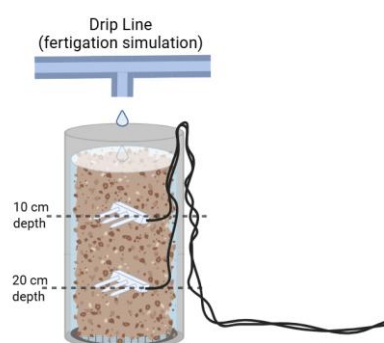


Figure 3. Column study—in lab fertigation simulation.

Table 2. Characteristics of soils.

Soil Type	Sand (%)	Silt (%)	Clay (%)	pH	Lime Index	BrayP1 P	K (ppm)	Ca (ppm)	Mg (ppm)	NO ₃ (ppm)	OM (%)
Sand	91	2.9	6.1	4.6	72	38	6	43	4	2.1	0.3
Loamy sand	85.5	4.9	9.6	4.1	65	57	16	93	11	12.3	4.3
Sandy loam	77.6	11.9	10.5	6.7	72	73	66	607	56	4.1	0.9

2.3.2. Field Experiment

A fertigation experiment was conducted at Michigan State University's West Central Michigan Horticulture Research and Extension Center (WCMHREC) in Oceana County, Michigan, to track and monitor the fertilizers through EC readings by TEROS-12 sensor. This field has sandy loam soil. The characteristics of soil in this field are shown in Table 3. Based on the survey, farmers practiced fertigation for high-density apple orchards in similar pattern as reported in Section 2.2. In an already established apple orchard with four rows (consisting of 11 panels per row and 12 trees per panel) planted 2.5 feet apart in a row, 15 panels (comprising of 180 trees) were randomly selected for this fertigation trail. This random selection was used to avoid bias and generalize results broadly to similar orchard systems. The fertigation solution was applied through a carefully calibrated fertigation system with ½-inch drip irrigation tape containing 2 L hr⁻¹ emitters spaced 24 inches apart. The fertigation solution was delivered to the trees via a DEMMA MixRite TF-5 model 005 liquid fertilizer injector. This injector was adjusted to 3% flow rate based on the total volume of irrigation allowing consistent delivery across all experimental units. An overview of the apple field and fertigation system is represented in Figure 4. Fertigation was applied for a 60 min period, followed by a 15 min irrigation cycle with clean water to prevent potential damage of drip lines caused by residual fertilizer solution. The clean water irrigation cycle was practiced ensuring longevity, reliability, and uniform application of irrigation system for replication and future use. The data were collected using Ubidots, a cloud-based IoT platform that enables real-time data visualization and analysis. This platform comprised a user-friendly interface that makes it easy for farmers to track the data in real-time and generate visual reports that enhanced the decision-making process based on trends, deviations, and/or issues in the fertigation system quickly.

Table 3. Characteristics of sandy loam soil at different depths.

Soil Depth	Sand (%)	Silt (%)	Clay (%)	Bulk Density (g/cm ³)	pH	Lime Index	BrayP1 P	K (ppm)	Ca (ppm)	Mg (ppm)	NO ₃ (ppm)	OM (%)
15 cm	72.6	15.9	11.5	1.356	5.9	69	244	155	789	170	13.0	3.2
30 cm	73.6	15.9	10.5	1.8748	5.0	68	168	35	228	66	4.9	0.9
60 cm	80.6	7.9	11.5	1.67	7.3	68	22	57	796	173	4.2	0.6

**Figure 4.** Overview of the (a) apple field and (b) fertigation system.

2.4. Data Analysis

The IoT-based EC and soil moisture data collected from the TERO-12 sensors were first calibrated against laboratory measurements to ensure accuracy and consistency. The calibration curve was generated to determine the correlation between sensor readings and standard EC values, with a high R^2 value confirming sensor reliability. After calibration, raw time-series data from the sensors were processed to remove noise and outliers using a rolling average method, as defined in Equations (2) and (3). This cleaning ensured that erratic spikes caused by environmental disturbances or brief connectivity issues were smoothed while retaining actual temporal trends. The cleaned dataset was then used to identify key fertigation events and to monitor changes in soil EC and moisture at different depths (10 cm and 20 cm), enabling temporal and spatial interpretation of solute movement and soil moisture.

GraphPad Prism software (10.5.0) was used to generate time-series plots for each treatment and soil type. These visualizations enabled the identification of peak EC values and moisture content before, during, and after fertigation. The differences between soil textures were examined based on their capacity to retain moisture and conduct ions, while fertilizer-specific EC responses were analyzed to evaluate nutrient mobility and potential leaching risk.

Overall, the data analysis workflow involved sensor calibration, signal preprocessing, temporal trend evaluation, and interpretation of soil–fertilizer–water interactions across treatments. This multi-step approach enabled a comprehensive evaluation of the IoT-monitoring system's ability to capture critical soil processes under fertigation regimes.

3. Results

3.1. Calibration of IoT-Based Sensors

Calibration of IoT-based sensors (particle reading) with YSI laboratory measurement of EC is shown in Figure 5. A strong correlation ($R^2 = 0.999$) between particle reading and YSI measurements for EC indicating the IoT-based sensor accuracy for real-time EC monitoring. The slope (1.0881) signifies that the IoT sensor slightly underestimates EC compared to the YSI, which may be attributed to differences in probe sensitivity or signal processing algorithms. The near-unity slope and minimal intercept (0.0247) suggest that the sensor is well calibrated and capable of delivering laboratory-comparable EC measurements with high precision. Such calibration practices are necessary to validate against the reference instruments for field deployment [22]. As suggested by recent study [24] ensuring a strong linear relationship through calibration enhances confidence in sensor performance under field conditions. This calibration ensures that the IoT-based sensor with particle reading provides reliable data of ions concentration in agricultural field and can be used in precision agriculture for continuous ion monitoring and decision-making.

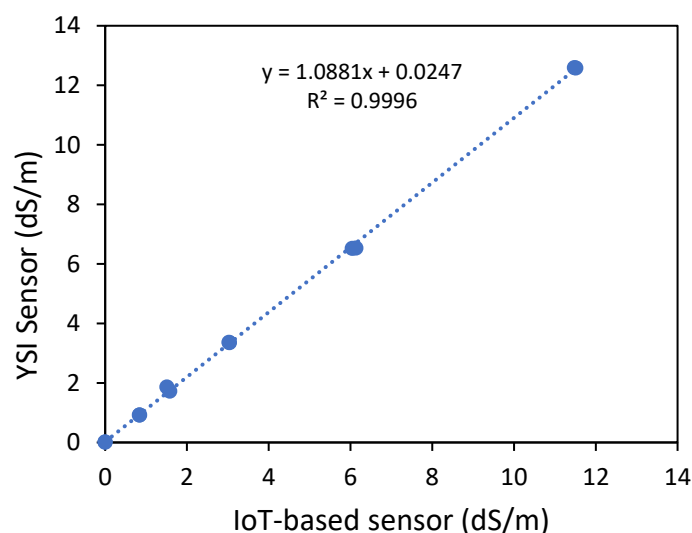


Figure 5. Calibration curve represents relationship between YSI laboratory measurement vs. IoT-based sensor measurement (particle readings) for EC.

3.2. IoT-Based EC and Soil-Moisture-Monitoring System—Column Study

3.2.1. Effect of Fertigation on EC and Soil Moisture in Sand

In sand, the response to fertigation was markedly different due to its coarse texture and high permeability. Figure 6 presents the dynamic response of EC and soil moisture contentment recorded by IoT sensors at two depths (10 cm and 20 cm) in a soil column over a short monitoring period. In Figure 6, EC shows a sharp initial increase at both depths, with the 10 cm sensor detecting a rapid spike followed by a near-immediate decline, while the 20 cm sensor shows a more sustained elevation before gradually tapering off. This behavior is indicative of solute infiltration dynamics; as water containing dissolved ions percolates downward, it first reaches the upper sensor, then the lower one. The faster peak and shorter duration at 10 cm suggest rapid initial infiltration, possibly influenced by capillary flow and higher concentrations near the surface. The extended EC peak at 20 cm indicates slower ionic migration and dispersion through the soil matrix. This reflects the poor buffering capacity and low ionic exchange potential typical of sandy soils [25]. Nocco et al. (2019) in their study also suggested that sand shows narrow range of EC values [26].

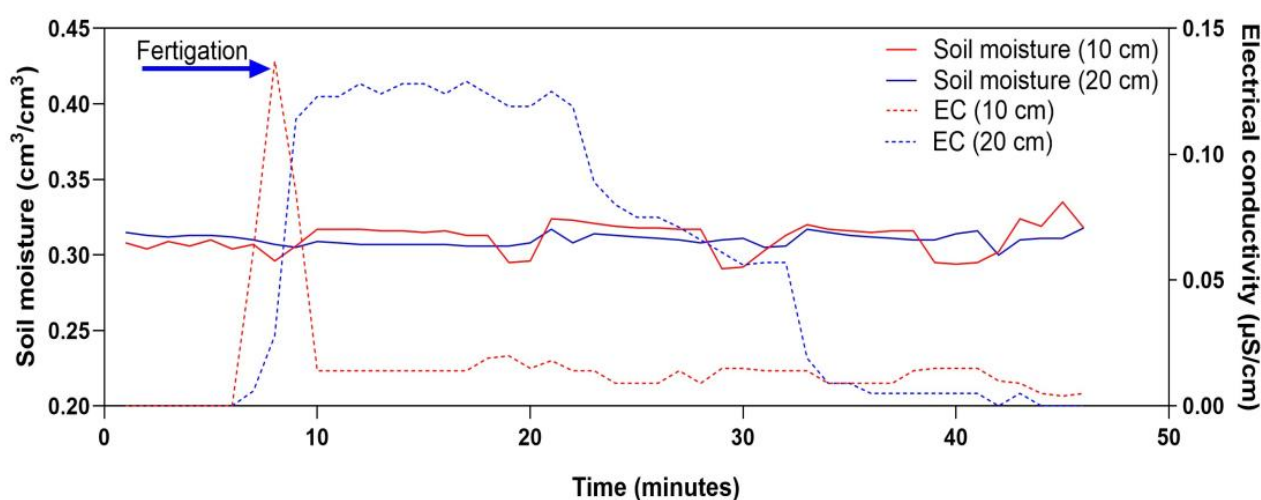


Figure 6. Soil moisture (cm³/cm³) and EC (µS/cm) data from IoT-based sensor for 10 cm and 20 cm depths in sand with an arrow showing fertigation application.

The soil moisture data (Figure 6) remained relatively stable at both depths, fluctuating slightly between 0.30 and 0.34 cm³/cm³. The near-simultaneous response at 10 cm and 20 cm suggests rapid vertical movement of water with limited storage. The limited moisture retention reduces the residence time of nutrients, increasing the risk of leaching losses and reducing nutrient-use efficiency [27].

3.2.2. Effect of Fertigation on EC and Soil Moisture in Loamy Sand

Soil moisture content and EC dynamics for loamy sand are shown in Figure 7. The data reveal distinct patterns in both water infiltration and solute transport due to the moderate texture and drainage properties of loamy sand [4]. The soil moisture curves indicate a clear wetting front, with a rapid increase in moisture at 10 cm starting around 75 min, followed by a delayed response at 20 cm occurring approximately 15–20 min later. The moisture content at 10 cm stabilizes at 0.31 cm³/cm³, while at 20 cm, it levels off near 0.21 cm³/cm³, reflecting the relatively higher water retention capacity near the surface and more limited percolation to deeper layers. This pattern is consistent with the typical hydraulic behavior of loamy sand, which retains water better than sandy soil but still exhibits moderate infiltration rates [28].

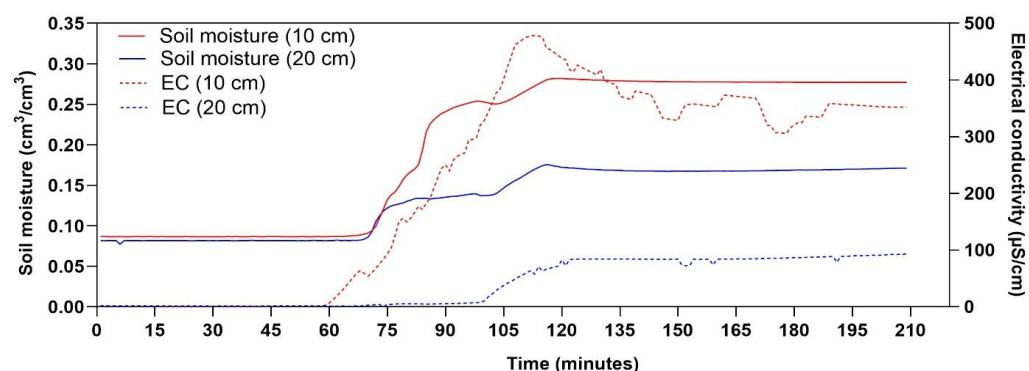


Figure 7. Soil moisture (cm³/cm³) and EC (µS/cm) at 10 cm and 20 cm depths in soil column with loamy sand.

The EC profiles further highlight the dynamics of the ion movement in response to fertigation. At 10 cm, EC rises sharply after 75 min, reaching a peak of approximately 450 µS/cm around 110 min before gradually stabilizing near 370 µS/cm. This trend indicates rapid solute accumulation near the surface, driven by the dissolution of applied fertilizers. The 20 cm depth shows a slower and lower EC response, peaking at 160 µS/cm, suggesting delayed and attenuated solute migration likely due to ionic adsorption, dilution, or limited pore connectivity. The divergence in EC magnitude between the two depths reflects the depth-dependent mobility of solutes in loamy sand, with stronger retention in the upper layer.

The synchronous rise in soil moisture and EC at both depths confirms the strong coupling between water flow and solute transport under fertigation. The steeper EC gradient compared to moisture suggests higher ion concentrations in the surface zone, highlighting the need for careful fertigation management to avoid salt accumulation and nutrient leaching. These observations support the value of integrated EC–moisture monitoring for optimizing irrigation and nutrient scheduling in textured soils like loamy sand.

3.2.3. Effect of Fertigation on EC and Soil Moisture in Sandy Loam Soil

The fertigation event in sandy loam soil produced distinct temporal responses in both EC and soil moisture. Figure 8 presents the EC and soil moisture data collected from IoT-based sensors placed at 10 cm and 20 cm soil depths over a monitoring period

of approximately 110 h. The figure provides a clearer and interpretable EC dynamics representation with time. The EC trends clearly depict an initial sharp increase followed by a gradual decline in EC at both depths, eventually reaching a relatively stable plateau. This behavior corresponds to the infiltration and redistribution phase of solutes through the soil profile [4]. Interestingly, EC values at 10 cm show higher initial peaks than those at 20 cm, suggesting earlier and more intense ionic activity near the surface following water or solute application. Over time, the convergence of EC values between the two depths indicates equilibration and vertical migration of solutes [29].

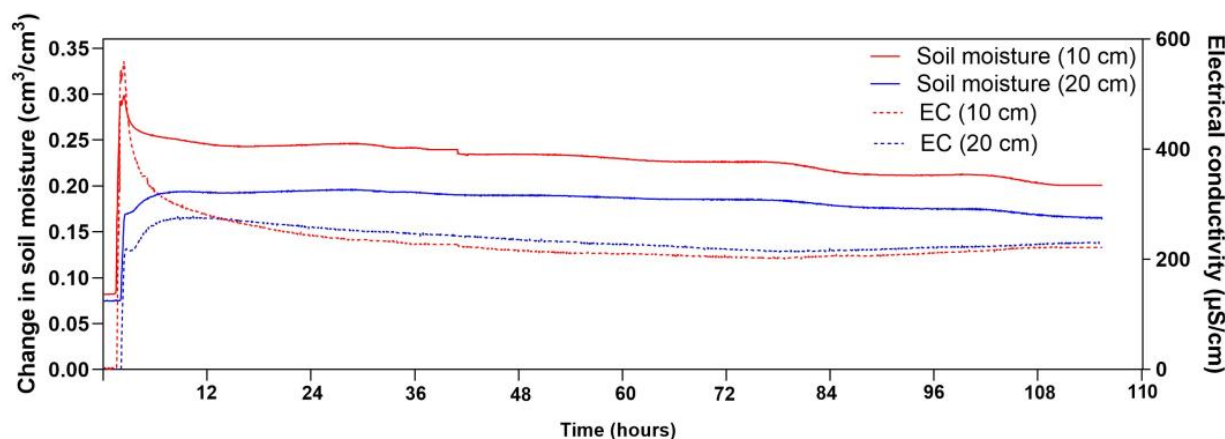


Figure 8. Electrical conductivity ($\mu\text{S}/\text{cm}$) and soil moisture (cm^3/cm^3) data from IoT-based sensors for 10 cm and 20 cm depths in soil (sandy loam) columns showing fluctuations over time.

Figure 8 illustrates the corresponding soil moisture dynamics. As with EC, the upper layer (10 cm dept) exhibits a faster and higher increase in moisture compared to the 20 cm depth due to direct exposure to infiltration. The moisture retention in the topsoil remains higher throughout the duration, while the 20 cm profile shows relatively stable but lower values, indicative of reduced infiltration or greater gravitational drainage. The synchronization between moisture and EC peaks indicates that ionic transport is tightly coupled with water movement.

3.2.4. Differential EC Response to Fertilizer Applications in Soil Columns

Soil EC is influenced by different ions present in soil, which depend on its type and behavior [30]. The EC (dS/m) response at 10 and 20 cm depths in soil columns treated with Calcium Ammonium Nitrate (CAN) and Monoammonium Phosphate (MAP) fertilizers along with control (water only) application is represented in Figure 9. The results clearly demonstrate the influence of fertilizer type and placement depth on ion mobility and solute distribution in the soil profile. The EC trends showed a clear increase for under CAN application compared to control treatments, which showed a stable trend for EC. The EC increased steadily after fertigation, remained plateaued, and then decreased with time due to water and nutrients movement in soil. Across depth, the increase in EC was prominent at the 10 cm depth, showing immediate solutes concentration in upper soil layer. This sharp increase is attributed to the high solubility and ionic strength of CAN, which rapidly dissociates into nitrate and ammonium ions, significantly raising the soil solution's conductivity [8]. The lower soil layer showed a slow and less pronounced increase in EC indicating downward movement of nutrients across the soil profile. CAN application resulted in higher EC due to nitrate ions, whereas MAP-applied soil column EC showed a decrease after fertigation due to low mobility of phosphorus ions. Post-fertigation of MAP showed a slight decrease in EC with time across the depths. Control with water application only showed a stable trend for both soil layers. This suggests a more buffered or complex

interaction between MAP components and soil chemistry. The phosphate component may have been adsorbed or immobilized in the upper soil layer, limiting its vertical migration. At 20 cm, the MAP-induced EC increase was subtle and stabilized quickly, suggesting limited leaching potential under these conditions.

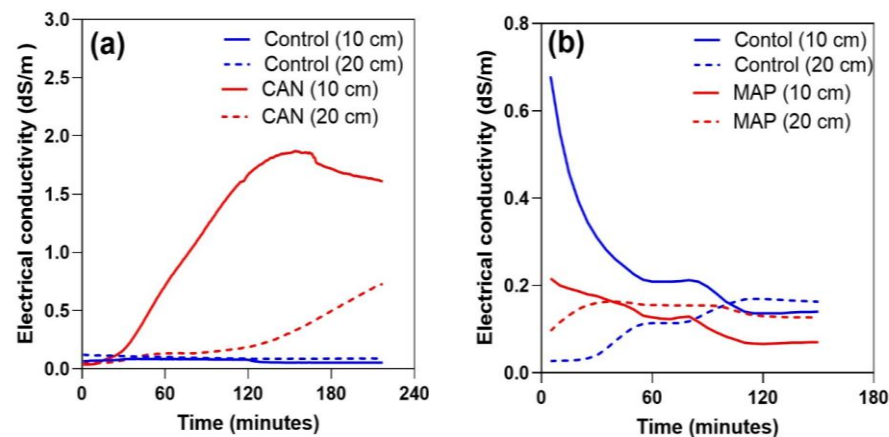


Figure 9. EC (dS/m) response at 10 and 20 cm depths in soil columns treated with (a) CAN and (b) MAP fertilizers along with control (water only) application.

These observations suggest that EC measurements can effectively differentiate fertilizer behavior in the soil and help predict their agronomic efficiency. Fertilizers with high leaching tendencies, such as CAN, may require split applications or lower doses, while more stable formulations like MAP can be strategically used for sustained nutrient delivery. Thus, continuous EC monitoring offers a practical, real-time approach for optimizing fertilizer management based on nutrient mobility and soil depth interactions.

3.3. IoT-Based EC and Soil-Moisture-Monitoring System—Field Study

Distinct fluctuations in soil moisture contents and EC in open-field conditions under irrigation and/or fertigation events are represented in Figure 10. The soil moisture profile exhibited distinct peak-and-decline cycles corresponding to irrigation events, with higher moisture content immediately following water application, followed by a gradual decrease due to infiltration, drainage, and evapotranspiration [31]. Early in the monitoring period (0–25 days), the peaks were more pronounced and frequent, indicating consistent irrigation inputs. However, as time progressed, a declining trend in both peak amplitude and baseline moisture levels was observed, possibly due to increased crop water demand, the drying of the soil profile, or reduced irrigation efficiency [32]. In parallel, EC values showed rapid spikes following fertigation events, notably around days 5, 10, 20, and 50, reflecting the influx of dissolved nutrients or salts into the root zone. These EC peaks were often followed by sharp declines, suggesting active leaching or dilution of solutes through the soil matrix [33]. Notably, the largest EC increase occurred near day 50, which is likely to correspond to a high-concentration fertigation event. The synchronized fluctuations in EC and soil moisture underscore the strong coupling between water and solute dynamics under field conditions [21]. These trends demonstrate the capability of real-time sensor networks to detect fertigation impacts and identify potential nutrient-leaching risks.

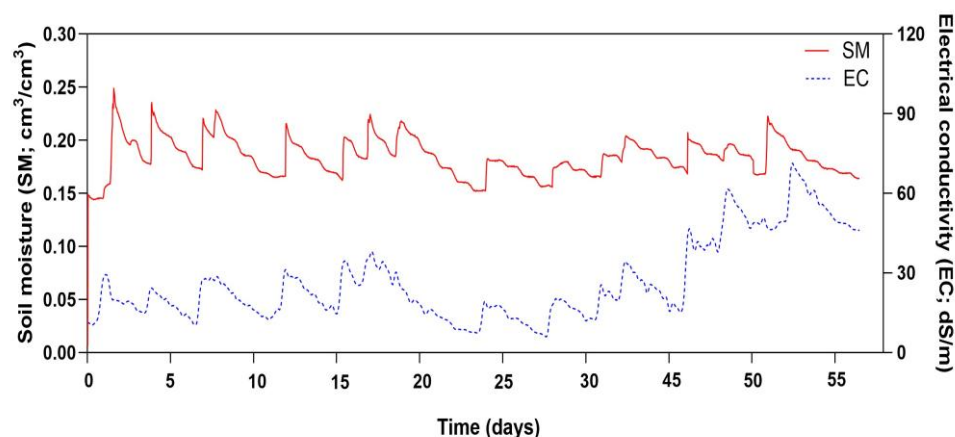


Figure 10. Time-series soil moisture contents (cm^3/cm^3) and EC (dS/m) in open-field representing peaks for irrigation and/or fertigation events.

3.4. Practical Implications of IoT-Based EC and Soil Moisture Sensors in Fertigation Management

The results of this study demonstrate the value of integrating IoT-enabled sensors for real-time monitoring of EC and soil moisture in the context of fertigation management. Real-time measurements of these parameters are critical for improving nutrient-use efficiency and reducing leaching losses, particularly in coarse-textured soils where nutrient mobility is high [15,20]. The IoT-based soil-monitoring system employed in this study exhibits several key advancements over conventional and recently reported monitoring technologies. Traditional systems often rely on manual sampling or data loggers lacking real-time connectivity, resulting in delayed detection of fertigation anomalies and inefficient nutrient management [18,19]. In contrast, the integrated use of Particle Argon microcontrollers and TEROS-12 sensors enables continuous measurement of soil EC and moisture at 15 min intervals, with data transmitted via Wi-Fi to a cloud-based platform (Ubidots) for immediate visualization and decision support. Moreover, many existing EC and moisture-monitoring systems rely on single-parameter sensing or inflexible analog designs that lack integration with IoT platforms [21,29]. The current system adopts a SDI-12 communication protocol for multi-sensor integration and supports modular deployment, making it well suited for both experimental studies and scalable field applications. The system also operates on a solar-powered platform with local data backup, ensuring autonomous function even under unreliable network conditions.

The observed synchronization between fertigation events and EC peaks confirms the sensor system's ability to detect fertilizer application timing and subsequent ion migration through the soil profile. This enables users to make informed decisions regarding fertilizer dose and application frequency. For example, the system clearly captured nitrate-driven EC spikes following CAN application, indicating higher solute mobility in contrast to the more stable EC profiles observed for MAP, which exhibits limited phosphorus transport due to strong sorption in most soil types [7].

Simultaneous moisture data allow for precise assessment of soil water status, which directly influences solute transport, particularly under drip irrigation scenarios. Rapid infiltration and EC dilution observed in sandy soils, as recorded by the IoT sensors, underscore the importance of high-frequency, depth-resolved monitoring to mitigate nutrient losses in permeable soils [4]. The sensors' responsiveness to both moisture and EC fluctuations suggests their suitability for adaptive fertigation control based on soil feedback.

From an operational standpoint, Ubidots enables remote data access, temporal trend visualization, and early detection of system anomalies such as incomplete fertigation or emitter failure. Users are recommended to perform periodic sensor calibration under site-specific conditions to maintain measurement accuracy, especially when used across

varying soil textures or salinity conditions. In summary, the deployment of IoT-enabled EC and moisture sensors can support precision fertigation by (i) identifying optimal fertigation timing based on real-time EC and soil moisture trends, (ii) detecting potential nutrient leaching events in sandy or shallow soils, (iii) enabling remote diagnostics of fertigation system performance, and (iv) supporting site-specific irrigation scheduling to reduce water and nutrient losses.

4. Conclusions

This study demonstrated the successful calibration and application of an IoT-based sensor system for real-time monitoring of soil EC and moisture under fertigation scenarios. The high correlation ($R^2 = 0.999$) between IoT sensor readings and laboratory-based YSI measurements confirmed the system's accuracy and suitability for precision agriculture applications. Temporal monitoring across soil columns revealed distinct responses in EC and moisture content depending on soil type and depth. Sandy loam exhibited the highest moisture and ion retention, with EC peaking at 550 $\mu\text{S}/\text{cm}$ and moisture stabilizing around 0.35 cm^3/cm^3 , indicating effective nutrient use and lower leaching risk. In contrast, sand showed rapid infiltration and lower EC (420 $\mu\text{S}/\text{cm}$), highlighting high nutrient loss potential. Loamy sand displayed intermediate behavior, with EC and moisture values reflecting moderate retention and solute mobility. Differential responses to CAN and MAP fertilizers further highlighted the importance of monitoring EC to assess nutrient availability and leaching potential, with CAN showing higher EC peaks and greater mobility than MAP. Field-scale validation confirmed the sensor network's ability to capture irrigation and fertigation events, reinforcing the value of integrated EC–moisture monitoring for adaptive nutrient and water management. Overall, the IoT-enabled monitoring system proved to be an efficient tool for enhancing fertigation efficiency, minimizing nutrient losses, and supporting site-specific management strategies tailored to soil texture and fertilizer type.

Although the system performed well under the tested conditions, limitations include the need for site-specific calibration, sensitivity to soil variability, and dependence on Wi-Fi connectivity. Future research should explore broader deployment across diverse field conditions, integration with low-power communication protocols, and coupling with automated fertigation control systems to support scalable, data-driven nutrient management strategies.

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